

Solar Panel Summative Project

Group 10

Ka Hin Cheung, Ethan Smith, James Matteucci, Alex Winter

Introduction

Project Aims

This report presents the feasibility of designing a deployable solar power generation system, mountable on a vehicle's roof. The company Tentbox, specialising in producing tents that mount to roof bars, wants to expand their product offering, thus the objective was to deliver a concise report which includes concepts generation, design development, motor and gear selection and other inputs which would lead to a well-designed, feasible product.

Stakeholders

Table 1 below shows a list of the different stakeholders for this product. The primary client was the consumer who held the values of sustainability and importantly cost and performance of the product. The secondary client was TentBox who were concerned mainly with costs and profits of the product. Another key stakeholder was the regulatory body which will constrain the dimensions and weight of roof mounted objects.

Table 1 – Stakeholders

Key Stakeholders	Description	Their Priorities
Company/Tentbox	Tentbox is the company which has contracted this project to expand their product offering. They consist of shareholders which are investors and are profit driven.	Cost, Sustainability, Profits
Consumer	The consumer base will consist of campers who often take road trips in the UK.	Safety, Cost, Performance, Sustainability
Regulators	These are regulatory bodies in the automotive industry which set out targets for sustainability and other measurement constraints	Safety, Sustainability
Design Team	The team which will be conducting the report on the feasibility of the project	Safety, Cost, Performance
Suppliers	Suppliers of standard components are raw materials in bulk	Use of standard of components
Manufacturers	These are the manufacturers which will manufacture the design of certain components for the product	Ease of manufacture, Cost

Background Research

Existing products showed the main issue with existing designs was the high aerodynamic effects of attaching a solar array to the roof of a vehicle (Sulaiman et al., 2018). This was an important point to note when the the PDS was generated and when the design concepts were made.

PDS

	Category	Type	Requirement	Target	Limit	Priority	Test
1	Requisites	Constraint	The mechanism must be wholly driven by a single 12V DC motor	—	—	Must	Model the system, producing a realistic simulation that will give required inputs from a 12V motor
2		Constraint	Deployment process is fully automated requiring no human intervention	—	—	Must	Sensors and dampers to detect when the cycle is complete so the mechanism stops at full deployment
3		Constraint	Design must employ only standard solar panels	—	—	Must	Research standard solar panel dimensions and design dimensions around this
4		Constraint	Design must be operable with a variable number of solar panels	0 < no. panels < max no. panels	—	Must	Design multiple frames that will hold solar panels separately
5	Size	Constraint	Width of retracted array must not exceed the width of the outer edges of a standard roof bar	< 110cm	120cm	Must	Measurements done on CAD
6		Constraint	Length of retracted array must not exceed the length of a car roof (dependent on car roof length)	< 130cm	140cm	Must	Measurements done on CAD
7		Objective	Minimise the retracted height of array to ensure aerodynamics	< 30cm	40cm	Should	Measurements done on CAD

8	Safety	Constraint	Mechanism must have sensors to automatically shut off the system in case of malfunction or misuse	—	—	Must	Automation report discussing sensors and how are implemented to ensure auto emergency stop
9		Constraint	All edges of mechanism should be filleted to ensure no sharp edges	—	—	Must	Fillet tool used on CAD
10	Durability	Constraint	Mechanism must reliably carry out a deployment cycle 10 times with no visible change in performance	n = 15	n = 10	Must	Tested when a prototype is made
11		Constraint	Mechanism must be secure and not deform under an added 10kg load in the deployed position	n = 15kg	n = 10kg	Must	Tested when a prototype is made
12		Constraint	Array must have a level of corrosion resistance to operate in an outdoor environment	C3 (ISO12944-2)	C3	Must	Selecting a suitably corrosion resistant material for the mechanism
13	Performance and Efficiency	Objective	Minimise total mass of design to reduce fuel consumption	< 50kg	80kg	Should	Calculations made based of materials used and their material densities as well as accounting for masses of standard components
14		Objective	Maximise surface area of deployed solar array by ensuring a high SA ratio between deployed and retracted positions	Ratio: 2:1	1.5:1	Should	Iterate and assess multiple design concepts
15		Objective	Minimise power consumption during deployment by minimising mass	Mass < 50kg	80kg	Should	Calculations based off torque required and time of deployment can help calculate power consumption
16		Objective	Maximise efficiency in the gearbox to reduce power consumption	Efficiency > 75%	η = 60%	Should	Calculations based off type and number of gears used and their efficiencies
17		Objective	Maximise motor efficiency by selecting an appropriate motor for our system's torque	Efficiency > 35%	η = 30%	Should	Use python to produce efficiency graphs for different motors at different torques
18		Objective	Optimise pitch angle for maximum absorbtion in the UK	Angle +/- 3°	+/- 5°	Should	Compare true angle with the optimal sun angles
19	Manufacture	Objective	Minimise number of components required to be machined/made	n < 30	n = 50	Should	Choosing simple designs
20		Objective	Use as many standard components as possible, requiring fewer manufacturing processes	—	—	Should	Using knowledge of extisting standard components and designing with these in mind
21		Objective	Design with viable, cost effective manufacturing methods	—	—	Should	Check each manufacturing process available at each step of design
22	Ease of use	Objective	Minimise deployment time	T < 25s	40s	Should	Choosing an appropriate motor by comparing the motors' various deployment times with models on Python
23		Objective	Minimise installation difficulty for client by reducing number of parts and ensuring a simple design	n < 30	n = 50	Should	Iterate and assess multiple design concepts
24		Objective	Provide a simple installation guide for panels with as few steps as possible	—	—	Could	Tested when a prototype is made
25	Aesthetics	Objective	Mechanism could be aesthetically pleasing	—	—	Should	Design concepts assessed and aesthetics taken into account
26	Cost	Objective	Mechanism should not be excessively expensive	Total cost < £50	£75	Should	Cost analysis for manufacture of final design assessed against target

Design Concepts

Following the identification of constraints and certain targets in the PDS, a number of design concepts were generated holding these values in mind.

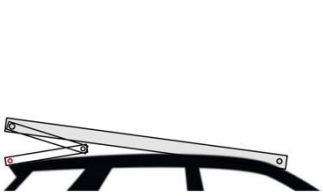


Figure 1a – Retracted (side view)

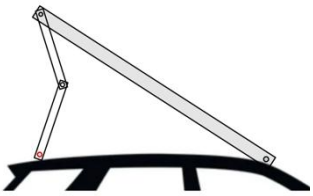


Figure 1b – Deployed (side view)

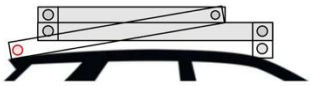


Figure 2a – Retracted

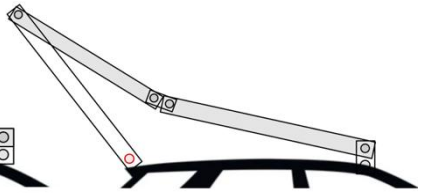


Figure 2b – Deployed

Design Concept 1

The first design displayed in figures 1a and 1b had the lowest complexity being a 3-bar linkage array driven by a single motor (indicated by the red circle). Due to its simplicity this design was designated as the datum. The shaded bar indicates where the solar array would be housed.

Design Concept 2

Design Concept 2 displayed in figures 2a and 2b represented a higher complexity linkage array with increased solar surface area when deployed.

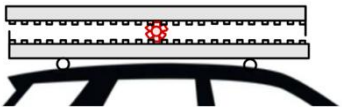


Figure 3a – Retracted

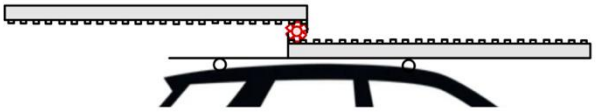


Figure 3b – Deployed



Figure 4a – Retracted

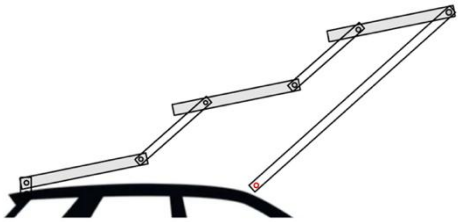


Figure 4b – Deployed

Design Concept 3

Design 3 (Figures 3a and 3b) used a rack and pinion to deploy the array, employing linear motion to double the surface area. The downside to this design was the inability to pitch the panels at an angle.

Design Concept 4

The fourth design (Figures 4a and 4b) use 3 arrays that fold out over the vehicle, prioritising the increase in surface area over the aerodynamic when retracted.

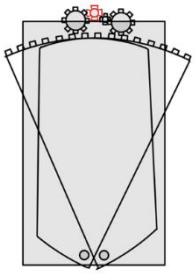


Figure 5a – Retracted

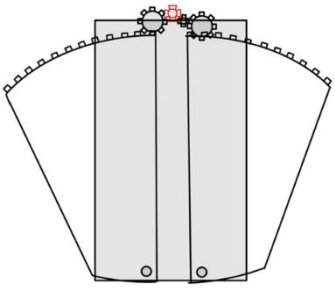


Figure 5b – Deployed

Design Concept 5

The final design (Figures 5a and 5b) shows 3 arrays that fold out like wings on a rotary rack and pinion. The downsides of this design being the inability to have pitch it and the irregular shape of the wings that the rectangular solar panels would fit to.

Convergence Table

Table2 – Convergence Table for Best Design

		Scores				
Criteria	**Weighted Values**	**Design 1 (Datum)**	**Design 2**	**Design 3**	**Design 4**	**Design 5**
Retracted Height	1	0	0	0	-2	-1
Deployed Surface Area	1	0	2	1	1	2
Number of Parts (Minimise)	0.25	0	-1	0	-2	-2
Deployment Time (Minimise)	0.25	0	-1	0	0	-1
Mass (Minimise)	1	0	-1	-2	-1	-2
Pitch Angle	0.75	0	0	-2	0	-2
Ease of Manufacturing	0.5	0	0	0	-1	-2
Safety	0.5	0	0	-1	0	1
Cost	0.5	0	0	-1	-2	-2
Structural Integrity/ Strength	0.75	0	1	1	0	1
Aesthetics (Subjective)	0.5	0	1	0	1	1
Weighted Score		0	1.75	-2.75	-3.5	-3.5
Rank		**2**	**1**	**3**	**5**	**5**

The convergence table, shown below in Table 2, compared the design concepts to some of the criteria from the PDS and aimed to ascertain which design met the PDS most effectively. The criteria were assigned weightings based on their importance and the designs were scored based on how effectively they met each criterion.

Weighted Values

Retracted height, deployed surface area and mass were all given a weighting of 1 as they were the key objectives from the PDS all referring to the performance and drag effects that were of a high priority. Pitch angle and structural integrity were the next highest weights as they also directly affect performance and safety. Ease of manufacturing safety and cost were weighted at 0.5 as these were some key points for the stakeholders and finally, number of parts and deployment time were deemed as the least important as the designs did not show clearly the results of this category.

Selected Design

The selected design, design 2, was ranked the highest in the convergence table with the highest weighted score of 1.75. This design gave multiple pitch angles for sun absorption at different times of the day, doubled deployed surface area, remained flat to the roof in its retracted position and kept structural integrity. Whilst it may not have had the highest deployment surface area ratio as discussed in the PDS, it excelled in multiple high priority PDS points making it the highest ranked design.

Design 2 was then prototyped using Lego for a physical model of the design mechanism as seen in Figure 6a and 6b. This allowed for the mechanism to be physically tested and ensure the deployment was truly viable. The design was also iterated with the longer pieces of Lego to maximise the surface area of the panels, improving on the initial concept.

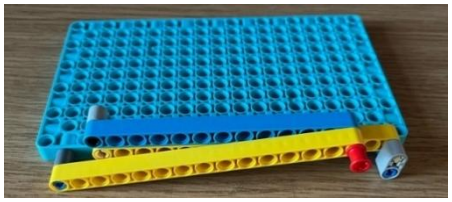


Figure 6a – Lego Prototype (Retracted)

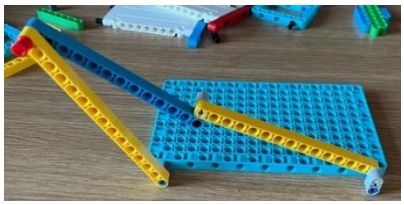


Figure 6b – Lego Prototype (Deployed)

Design Iteration

UK Sun Positioning

The angles of design two's array were chosen based on research conducted on the sun positioning in the UK during the year to hit PDS requirement 18. The sun angles were plotted onto a graph as seen below in Figure 7a, with the use of the pvlib library to conduct an accurate year-round iridescence analysis focusing on months between May to September as these were the most common months for consumers to camp (The Expert Camper). The figures below show the first optimal angle to be 25°, especially during peak iridescence, whilst the second angle 15° tilt (close to the final 16° tilt) was more favourable during the month of May. The decision of using 25° tilt was further reinforced by Figure 7b proving 25 degrees to be most optimal.

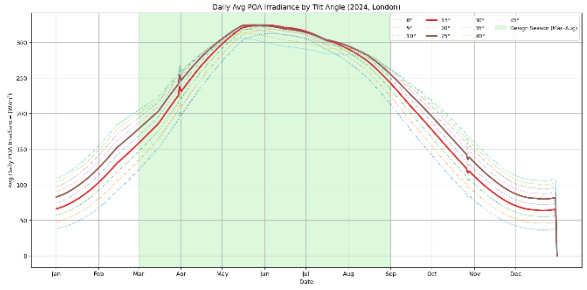


Figure 7a – Energy Produced vs Months of the Year (Varying Tilt) Angles)

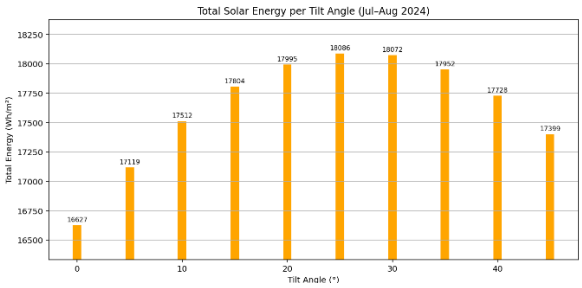


Figure 7b – Total Solar Energy Produced for Varying Tilt

Dimensional Constraints

As there is no standard size of roof rack and the dimensions for these being variable between vehicle manufacturer, the most sold SUV (most common type of vehicle used by the consumer) was selected as a base. According to the RAC the most sold car of 2024 (48,340 units) was the Ford Puma (RAC Forum, 2024), with roof dimensions of 1.2m x 1.4m. The standard solar panel selected was the VSSP-20/12 which measured 440x350x25mm.

Material Selection

The part acts as a linkage member therefore it can be expected to encounter a significant torque such that material strength will drive selection. This is constrained by PDS items 10 and 11. Additional quantities to be considered include minimising cost and weight as well as being corrosion resistant (class C3) as constrained in PDS items 13,26 and 12. Yield strength will be used to compare resistance to plastic deformation due to torque. The resulting Material Index is shown below in equation [1].

$$M = \frac{\sigma}{\rho C} \quad [1]$$

Where σ represents yield strength in MPa and ρC represents cost and density in g/cm³. Several suitable materials were plotted between these axes in figure 2 below such that more suitable materials are found in the top left quadrant of the plot.

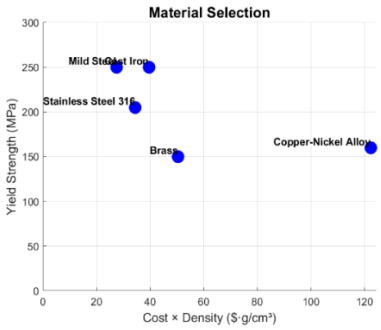


Figure 8 – Material Selection Chart

Ultimately 316 Stainless Steel was selected as the material to produce this part. Additional factors governing this decision included: exceptionally high corrosion resistance; suitability for a large range of manufacturing processes as defined in PDS item 21; suitability for a spectrum of surface finishing processes; a low thermal expansion coefficient that reduces the likelihood of expansion and contraction with temperature that can loosen or damage fixings; and relative ease to source when compared to other steel alloys.

Initial CAD and Design Iterations



Figure 9 – Initial Retracted Design

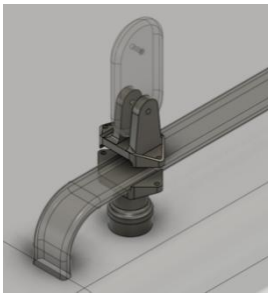


Figure 10 – Initial Roof Bar Attachment

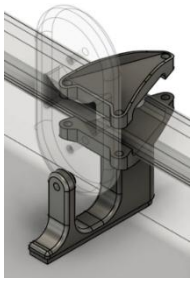


Figure 11 – Roof Bar Attachment Iterated

Having selected design concept 2, an initial model shown in Figure 9 was produced (relevant PDS items: 5,6,7,19,20,25). A critical analysis of this design identified its primary drawbacks as being excessive mass resulting in high torques applied to the motor, a large distance from the bottom of the solar array to the roof of the car thereby enlarging the profile of the car potentially worsening fuel consumption through additional drag as well as introducing aesthetic concerns. Shown in Figure 10 is the preliminary method of attachment, designed prioritising user operability as well as its function as a means by which the solar array remains rigidly fixed to the roof bars during regular use (PDS) and during the opening and closing of the linkage array. The design required the use of two M6 bolts and two M6 nuts to secure the two halves of the part to the roof bar, while the bottom half itself was supported by the car roof. Here the linkage array sat on top of the roof bar contributing to the height concerns of the initial design. The small profile of the bottom half of the attachment may also result in high pressure on the car roof and thus future iterations should increase the area over which this force is spread.

The iterated attachment shown in Figure 11 Addresses the height concern in having the array attached on the inner edge of the roof bars, this corresponds to a reduction in height of 85mm or a percentage decrease of 38.6%. Additionally, the larger footprint of the bottom half more evenly distributes the weight of the array thereby reducing the pressure on the car roof. The use of a 3rd M6 nut and bolt improves rigidity reducing the likelihood of vibration transmission from the car which may negatively impact fixings. Figures 12 And 13 Show the final iteration of the solar array in both a retracted and opened positions, the linkage bar members themselves were iterated as set out in the manufacturing report.

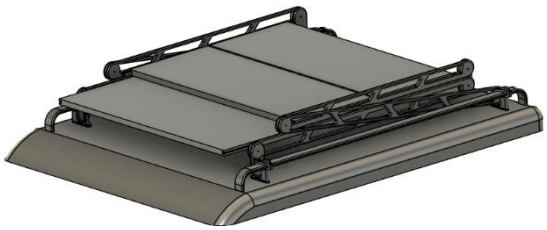


Figure 12 – Iterated Retracted Design



Figure 13 – Iterated Deployed Design

Fixings

Fixing selection was influenced by the requirement for ease of assembly/disassembly and maintenance. (relevant PDS items: 20,23,26). Additionally, complexity was minimised where possible therefore fixing arrangements should minimise number of parts. Fixings were required to allow rotational motion between linkage members while allowing the mechanism to maintain its structural integrity. Product lifespan was also considered and these factors allowed for the selection of a fixing arrangement between linkage members. Each fixing arrangement will therefore contain two RS PRO SMF106ZZS Single Row Deep Groove Ball Bearings, here shielded ball bearings were selected to improve product lifespan. The bearings will be held together with a M6 x 40mm Double Ended Threaded Bar whose smooth section would friction fit into the internal diameter of the bearings, the bearings and thus the linkage bars would be held together using two M6 dome nuts threaded onto either end of the threaded bar. This arrangement repeated 12 times for the entire assembly.

Motor Selection

Following the prerequisite requirement as outlined from PDS item 1, an electric 12V DC motor without transmission must be selected from the Bosch Motor Catalogue. To keep the system design relatively simple, motor selection was limited to just bidirectional motors, so a reverse transmission design is not required.

This saves on costs and time to design which links to PDS point 2 where the system should be fully automated. This reduced the number of motors left to consider to just four: APM, NSA-I P1, NSA-I P2, and NSA-I P3.

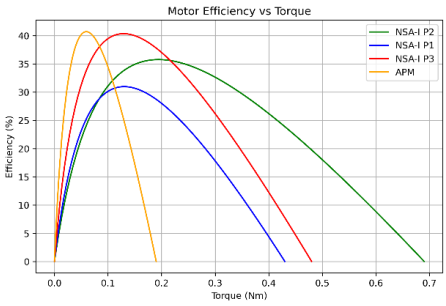


Figure 14 – Motor Efficiency vs Torque

Certain factors were considered to choose the optimal motor. One factor considered was the efficiency of each motor as operating torque varied, which was also referenced from PDS item 17 to consider motors with higher efficiencies. From the Bosch Motor Catalogue (Bosch, 2019), each motor had their own performance curves with specific information about how torque varies with angular speed and current. Using this information, a graph was plotted for each motor's efficiency. The full process is detailed in the technical analysis report. In figure 14, it is shown that APM and NSA-I P3 has the highest overall peaks in terms of motor efficiency at a low operating torque, and that NSA-I P1 has the overall lowest motor efficiency so this motor was ultimately out of consideration.

The next factor considered was the deployment times between each motor against varying gear ratios. PDS item 22 suggested a minimum time of 25s. The system was modelled as an inverted pendulum where all the mass is concentrated at a distance from the main pivot point (further detailed in the Technical Analysis Report). Each motor in the catalogue has details for its stall torque and no-load speed which could be imported into the python code which models the system. The following figures 15, 16, 17 and 18 were produced, which outline the deployment times taken for each motor at their most optimal gear ratio.

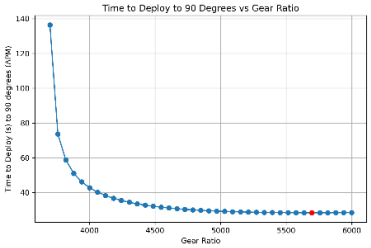


Figure 15 – Time Taken to Deploy (APM)

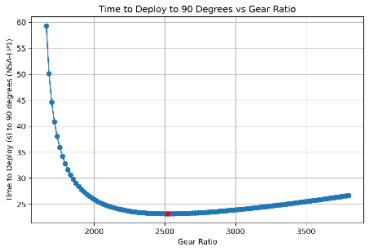


Figure 16 – Time Taken to Deploy (NSA-I P1)

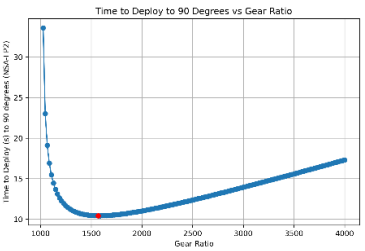


Figure 17 – Time Taken to Deploy (NSA-I P2)

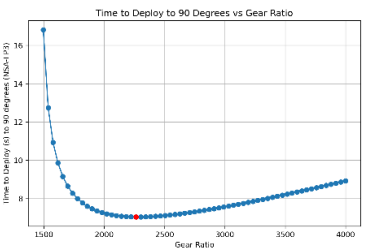


Figure 18 – Time Taken to Deploy (NSA-I P3)

As shown in the figures, APM motor requires a gear ratio of 5700 to deploy at the fastest time of 28.32 seconds, NSA-I P1 requires a gear ratio of 2520 to deploy at 23.18 seconds, NSA-I P2 requires a gear ratio of 1570 to deploy at 10.42 seconds and NSA-I P3 requires a gear ratio of 2263 to deploy at 7.05 seconds. From these figures, NSA- I P2 requires a reduced gear ratio to deploy the system within a suitable time range, whilst APM and NSA-I P1 takes a considerably longer time even at higher gear ratios. Minimising the gear ratio was optimal as this reduces the gearbox design's complexity, reduces any mechanical losses within the gearbox (linked to PDS item 16), save on manufacturing costs and reduce the stresses experienced within the gearbox.

Another factor was the degree of protection and the operating mode of each motor which is linked to PDS durability item 10 where the system should be able to deploy and retract 10 times without changes in performance. The APM motor had an operation mode of S1/S2, which was continuous duty and short-time duty respectively. Continuous duty runs at a constant load until thermal equilibrium and short-time duty runs at constant load for a limited time before stopping to cool down. This meant APM was not suitable for the system which will deploy and retract many times without proper time to

cool. NSA-I has an operation mode of S3 (intermittent periodic duty) which is characterised by reduced heat build-up and higher power output for intermittent loads. The NSA-I motors have an IP rating of IP 50 compared to the APM motor of IP rating 40, which means NSA-I has better dust resistance which will perform better in an outdoor environment (linked to PDS item 12).

Thus, the selected motor was NSA-I P2 because of its high torque which reduces the required gear ratio, relatively high motor efficiency which has a smaller gradient (such that it maintains a high efficiency even at higher torques) and suitable time of deployment. The optimal gear ratio selected was 1570.

Dampers

In order to select the optimal damper, the retraction phase of the system was modelled with damping ratios in mind (links to PDS point 2 and 10, where the system should be fully automatic and able to retract multiple times). Using the motor NSA-I P2's specification and a damping ratio of 1570, a graph where time taken to retract can be plotted against varying damping ratios (figure 19). To match the estimated deployment time of 10.42 seconds, as estimated value of 8000 Nms/rad for the damping ratio is most optimal.

This translates to a linear damping ratio of 32,000 Ns/m (analysis and method detailed in the technical analysis report), which allowed for the selection of linear dampers from the ENIDINE Catalogue (ENIDINE, 2023). The most suitable damper was the LD750 as its linear damping ratio was around 29,000 Ns/m which is within acceptable range to 32,000 Ns/m. It was found that the crank system which aims to convert rotational movement into linear movement should be joined at 0.5m away from the main pivot (figure 20) in order for the operational rotational speed to maintain a linear speed of at least 2.5 m/s (justified in technical analysis report).

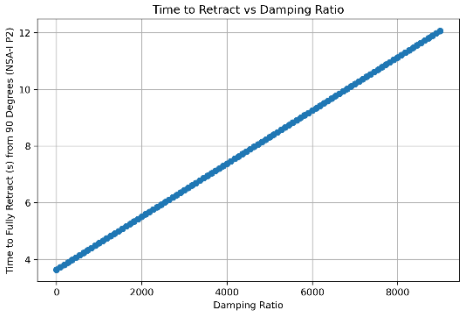


Figure 19 – Time Taken to Retract vs Damping Ratios

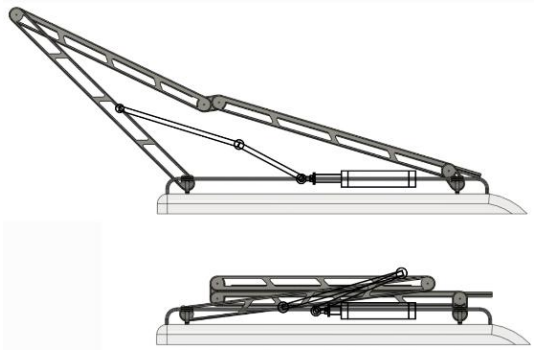


Figure 20 – Damper Placement

Gearbox Design

Having selected the motor NSA-I P2, with the optimal velocity ratio for the gearbox resulting being calculated as 1570, the gearbox could be designed. The first constraint on the gearbox was the 135mm clearance gap underneath the array that the gearbox would sit in. Next, to maximise efficiency, the target of no more than 6 step-ups was made. Within these 6 steps, a target ratio was chosen for each step-up between 1 to 6 to reduce gear diameter. After these initial targets and motor specifications were put into Table 3 shown below, the modules were estimated using the pinion speed power chart from blackboard (University of Bristol, 2024). To avoid gear hunting, the exact teeth numbers on the gears were selected, ensuring the highest common factor of each gear pair was 1, which gave a slightly different true velocity ratio of 1486. Next, the gear width was calculated in Table 3 using equation [3]:

$$W = 10M \quad [3]$$

Where W represents width and M represents module. Finally, each gear was then selected from the KHK catalogue as seen in Table 3 (KHK Gears).

Category		Meshing Gear No. 1,2	3,4	5,6	7,8	9,10	11,12
Power (Watts)		360	360	360	360	360	360
RPM	Pinion	3750	625	125	31.25	10.41667	5.208333
	Wheel	625	125	31.25	10.41667	5.208333	2.604167
Module		**1**	**1**	**1.5**	**2**	**3**	**3**
Target VR		6	5	4	3	2	2
No. Teeth	Pinion	17	17	17	17	17	17
	Wheel	110	84	66	48	35	35
Actual VR		6.470588	4.941176	3.882353	2.823529	2.058824	2.058824
Pitch Circle Diameter (mm	Pinion	17	17	25.5	34	51	51
	Wheel	110	84	99	96	105	105
Gear Width (mm)		10	10	15	20	30	30
Model No.	Pinion	SS1-17	SS1-17	SS1.5-17	SS2-17	SS3-17	SS3-17
	Wheel	SS1-110	SS1-84	SS1.5-66	SS2-48	SS3-35	SS3-35

Motor Values		**Units**
Target VR:	1570	
Max motor current:	30	Amps
Max motor torque:	0.69	Nm
Voltage:	12	Volts
Max motor RPM:	3750	rpm
Transmitted power:	360	Watts
Motor Torque:	69	Nmm
Actual VR:	1485.5925	

Table 3 – Gear Information

To test whether the module for each gear was truly suitable, each gear was tested for its maximum allowable load and compared against the true load through each gear. To calculate the maximum allowable load in each gear, the Lewis Form Factor graph shown in Figure 21 and equation [4] were used.

$$W_t = \frac{S\sigma FY}{P} \quad [4]$$

Where W_t represents tangential load, S represents the safety factor (3 was chosen), σ represents the tensile strength of the material (steel gears were used), F represents the face width of each gear, Y represents the Lewis Form Factor acquired from Figure 21, and P represents diametral pitch. These values were compared against the true load through each gear (calculated using equation [5]) to check the module for each gear was suitable and shown in Table 4.

$$W_t = \frac{2T}{d} \quad [5]$$

Where $W_t$represents tangential load, T represents torque and d represents pitch diameter.

Category		Meshing Gear No. 1,2	3,4	5,6	7,8	9,10	11,12
Diametral Pitch		1	1	0.666667	0.5	0.333333	0.333333
Bore Diameter (mm)	Pinion	8	8	8	12	15	15
	Wheel	10	10	15	15	20	20
Lewis Form Factor	Pinion	0.3	0.3	0.3	0.3	0.3	0.3
	Wheel	0.44	0.43	0.42	0.38	0.44	0.44
Maximum Allowable	Pinion	**500**	**500**	**1125**	**2000**	**4500**	**4500**
Tangential Force (N)	Wheel	**733.3333**	**716.6667**	**1575**	**2533.333**	**6600**	**6600**
Torque (Nmm)	Pinion	69	446.4706	2206.09	8564.82	24183.02	49788.57
	Wheel	446.4706	2206.09	8564.82	24183.02	49788.57	102505.9
Actual Tangential Force		**8.117647**	**52.52595**	**173.0267**	**503.8129**	**948.3538**	**1952.493**

Values		**Units**
Ultimate Tensile Stength (mild steel)	500	N/mm^2
Safety factor	3	
Allowable bending stress of mild steel	166.66667	N/mm^2

Table 4 – Force analysis on Gears

Once the gears had been specified and selected, the gearbox could be designed on CAD. First, the gears were imported to fusion from the supplier (KHK) and saved as components. Next using a compound gear train and the pitch diameters, the axles and housing were designed, helping align the gears as shown in figure 22.

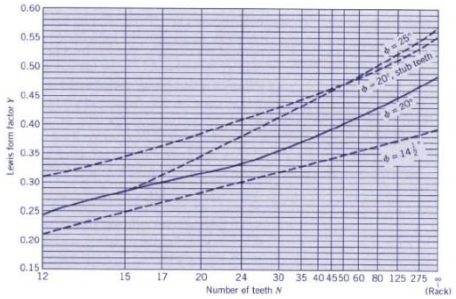


Figure 21 – Lewis Form Factor

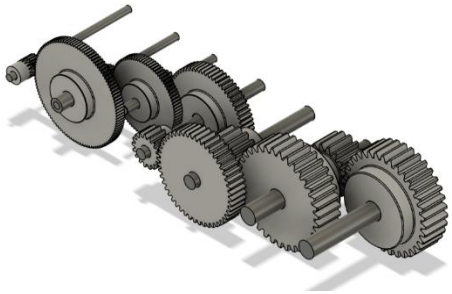


Figure 22 – CAD of Gear Train

Evaluation

Viability of Design

PDS requirement 7 expresses the importance of minimising the system's drag. A theoretical scenario was chosen to test the product's increased power consumption due to drag against the power produced in this scenario – a Londoner travelling to the Peak District for a week-long camping holiday. The system was modelled as a bluff body similar to a roof rack travelling at an average speed of 70 kph, giving an energy consumption due to the array's drag of 339.6Wh. Approximations and assumptions do not account for eddy forces forming between panels, therefore the worst-case scenario of the array increasing the coefficient of drag of the car by 40% (KTH Royal Institute of Technology, 2011) was considered. A maximum possible energy production of 3,080Wh was calculated using the values for energy production, considering the specifications of the VSPP-20/12 solar panels, assuming exposure for 7 days in peak iridescence. Contextually, 3 days of charge in these conditions would produce 1320Wh of energy, (approximately 1000Wh) equivalent to 110 full phone charges (EcoFlow, 2024) or 264 hours (11 days) of light from an LED lamp (U.S. Department of Energy, 2023).

This comparison shows the energy production to be much greater than the consumption due to drag. Naturally, the fewer days the array is deployed, the less power would be produced. Whilst these figures are satisfactory, best-case assumptions and large approximations were made to improve the feasibility of this product. The design could be modified (see figures 9 and 10) to improve aerodynamics and reduce drag, making trips of less time becoming viable.

Final Evaluation

The feasibility was evaluated against key targets in the PDS. Item 14 targets Retracted : Opened SA ratio of 2:1 which was exceeded at 2.5:1. Item 22 targets a deployment time less than 25 seconds which was exceeded at 10.42 seconds. Items 13 and 15 target a mass of under 80kg which was exceeded at 74.92kg.

Design Modifications

Following the economic viability assessments and informed by the PDS, further iterations were carried out aiming to improve efficiency and thereby reduce power consumption. With operability in mind, as well as ease of assembly/disassembly and weight minimisation.



Figure 23 – Retracted Design (Aerodynamic Attachment)



Figure 24 - Deployed Design (Aerodynamic Attachment) Attachment)

Figures 9 and 10 Show a proposed iteration in the form of an HDPE injection moulded shell, when closed the shell provides a more aerodynamic profile to aid in reducing drag, in order to allow the linkage mechanism to function the shell was divided into two parts with the top part attached to linkage 1 and the bottom part attached to linkage 3 by M6 dome nuts screwed onto the threaded bar. The masses of the top and bottom parts respectively are 4.10kg and 5.25kg which is negligible with the respect to the solar array. The parts serve a secondary function in protecting the solar panels from abrasive wear during driving thereby improve product life cycle.

Cost Breakdown

This section was to calculate the total cost of all components, both bespoke, standard or supplied parts. The aim was to gather a rough estimate of total cost of the array to assess the feasibility of the product and whether the product was of reasonable price for consumers.

Table 5- Cost Breakdown Table

Part Description	**Part Number**	**Supplier**	**Quantity**	**Per Unit Cost (£)**	**Total Cost (£)**
Gear 1	SS1-17	KHK Gears	2	4.29	8.58
Gear 2	SS1-110	KHK Gears	1	4.89	4.89
Gear 4	SS1-84	KHK Gears	1	5.39	5.39
Gear 5	SS1.5-17	KHK Gears	1	6.39	6.39
Gear 6	SS1.5-66	KHK Gears	1	6.89	6.89
Gear 7	SS2-17	KHK Gears	1	7.19	7.19
Gear 8	SS2-48	KHK Gears	1	7.39	7.39
Gear 9	SS3-17	KHK Gears	2	10.29	20.58
Gear 10	SS3-35	KHK Gears	2	10.59	21.18
Shielded ball bearing	SMF106ZZS	RS PRO	24	6.41	153.84
Double ended threaded bar	DIN 939	Accu	12	0.51	6.12
Dome Nuts	DIN 1587	Accu	24	0.03	0.72
Roof Bar attachment (Top)	Bespoke	Bespoke	4	0.11	0.44
Roof Bar attachment (Bottom)	Bespoke	Bespoke	4	0.02	0.08
Roof foot attachment	Bespoke	Bespoke	4	0.003	0.02
Linkage 1	Bespoke	Bespoke	2	7.12	14.24
Linkage 2	Bespoke	Bespoke	2	6.17	12.34
Linkage 3	Bespoke	Bespoke	2	7.12	14.24
Linkage 4	Bespoke	Bespoke	2	0.89	1.78
Linkage 5	Bespoke	Bespoke	2	0.89	1.78
				Total	**£294.08**

Table 5 is a list of components used in the solar array and gearbox and takes into account the labour costs of a continuous production line for the bespoke parts. Additional to these material and labour costs, set-up costs for machinery result in an array that costs around £300 to manufacture. For a consumer, also buying up to 15 of the VSSP-20/12 solar panels at £25.84 per unit would have an additional cost of £387.60.

BIBLIOGRAPHY

Bosch. (2019). *Electric Motors Catalogue 2019–2020*. [Catalogue] Bosch Rexroth AG. https://www.boschrexroth.com

EcoFlow. (2024). *What Can 1kWh Power?* [Online] https://www.ecoflow.com/blog/what-can-1-kwh-power

ENIDINE. (2023). *Seismic Products Catalogue*. [Catalogue] ITT Enidine Inc. https://www.enidine.com/en-US/

Engineers Edge. *Lewis Factor in Gear Design*. [Online] https://www.engineersedge.com/gears/lewis-factor.htm

KHK Gears. *KHK Gears Online Catalogue*. [Online] https://khkgears.net/new/

KTH Royal Institute of Technology. (2011). *Final Lecture: Drag and Lift*. [PDF] https://www.mech.kth.se/courses/5C1211/KTH_FinalLecture_11.pdf

RAC Forum. (2024). *The most popular cars of 2024 (so far)*. [Online] https://forum.rac.co.uk/threads/24355-The-most-popular-cars-of-2024-(so-far)

Sulaiman, S., Hassan, M.F., Yusoff, M.Z. and Rahim, M.S.M. (2018). *Aerodynamic Analysis of Solar Panel Mounted on a Car Roof Using Computational Fluid Dynamics (CFD)*. *International Journal of Applied Engineering Research*, 13(22), pp.15927–15932. https://www.ripublication.com/ijaer18/ijaerv13n22_03.pdf

The Expert Camper. *When is the best time to go camping?* [Online] https://theexpertcamper.co.uk/blog/camping/when-is-the-best-time-to-go-camping/

University of Bristol. (2024). *MECH20007: Engineering Practice – Lecture Materials*.University of Bristol Blackboard.

U.S. Department of Energy. (2023). *Estimating Appliance and Home Electronic Energy Use*. [Online] https://www.energy.gov/energysaver/estimating-appliance-and-home-electronic-energy-use